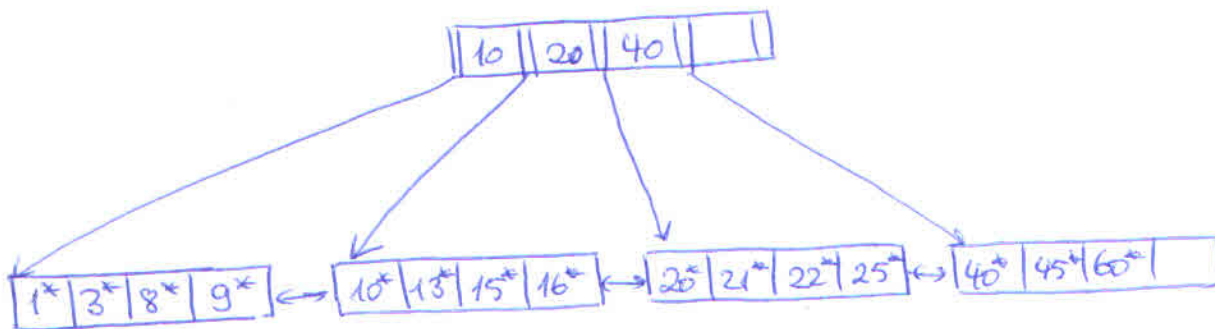


HOMEWORK #2 Solutions Set

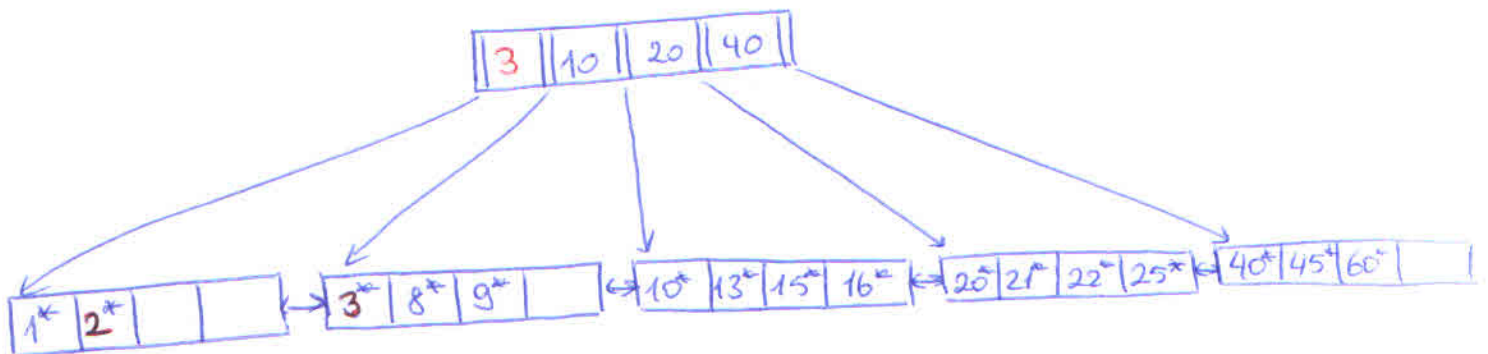
1. First, we need to sort the records wrt their keys.

1, 3, 8, 9, 10, 13, 15, 16, 20, 21, 22, 25, 40, 45, 60

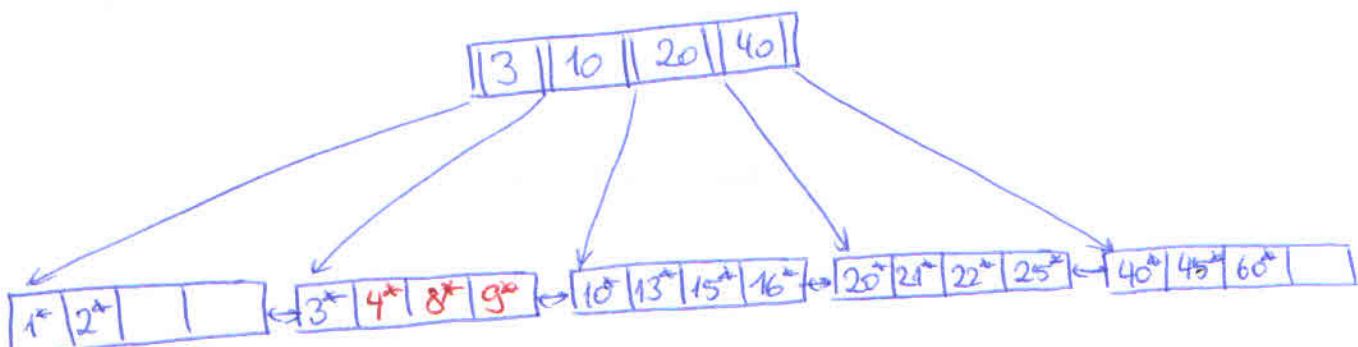
As $d=2$, Each node (except the root) will at least 2, at most 4 keys. The B+ tree after applying the bulk loading algorithm is given as follows:



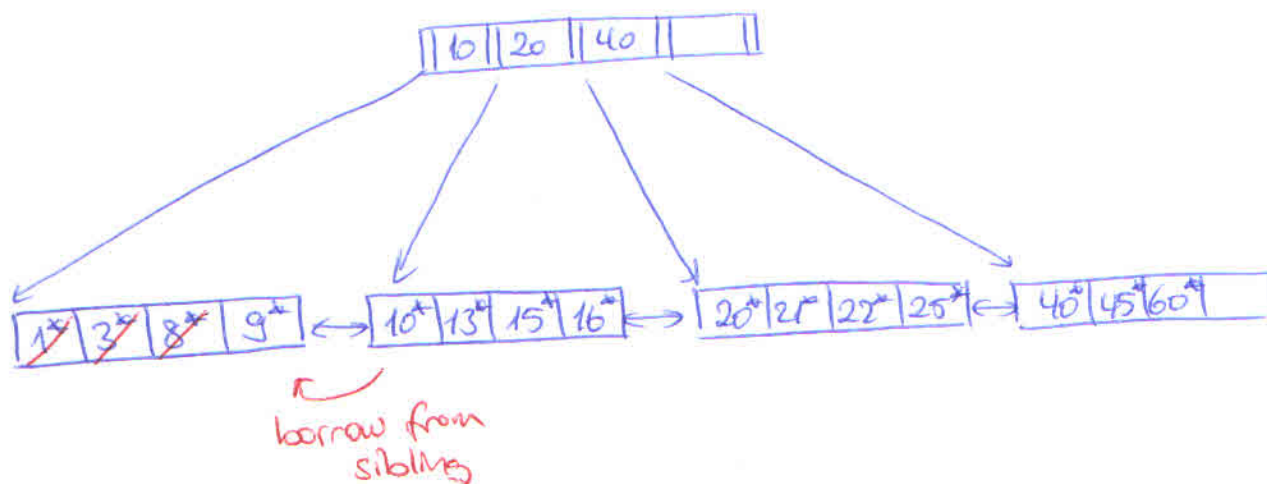
2. To insert 2, we need to split the first leaf node as below.



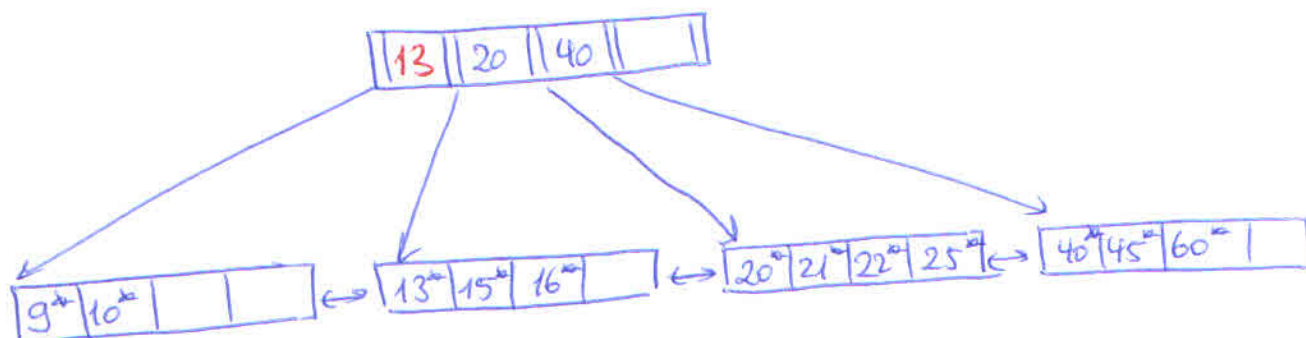
4 is inserted to 2nd leaf node as we have enough space.



3. Delete 1, 3, and 8 from the below tree:

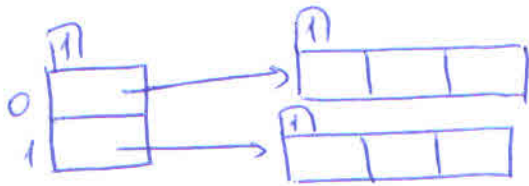


The final tree after deleting the records is as follows:



4.

Initially the extendible hash structure is as follows:



Insert the below values in the given order:

$$2 = (10)_2$$

$$15 = (1111)_2$$

$$23 = (10111)_2$$

$$6 = (110)_2$$

$$35 = (100011)_2$$

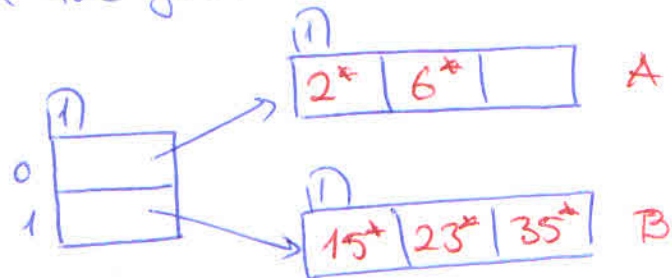
split $\rightarrow 13 = (1101)_2$

$$8 = (1000)_2$$

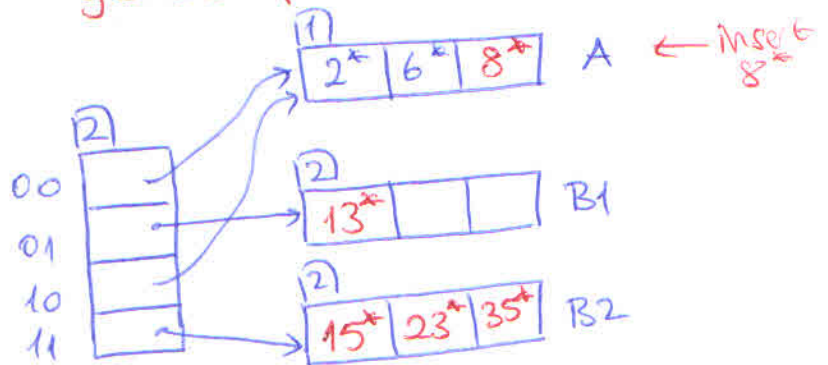
split $\rightarrow 26 = (11010)_2$

$$1 = (1)_2$$

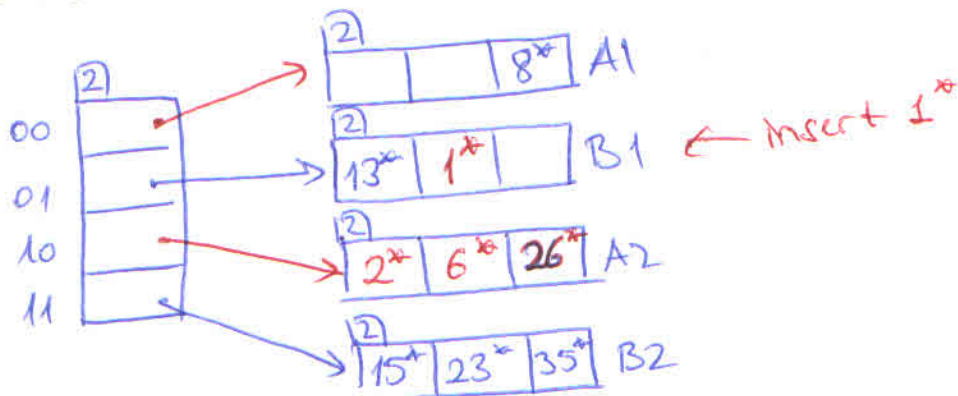
split $\rightarrow 11 = (1011)_2$



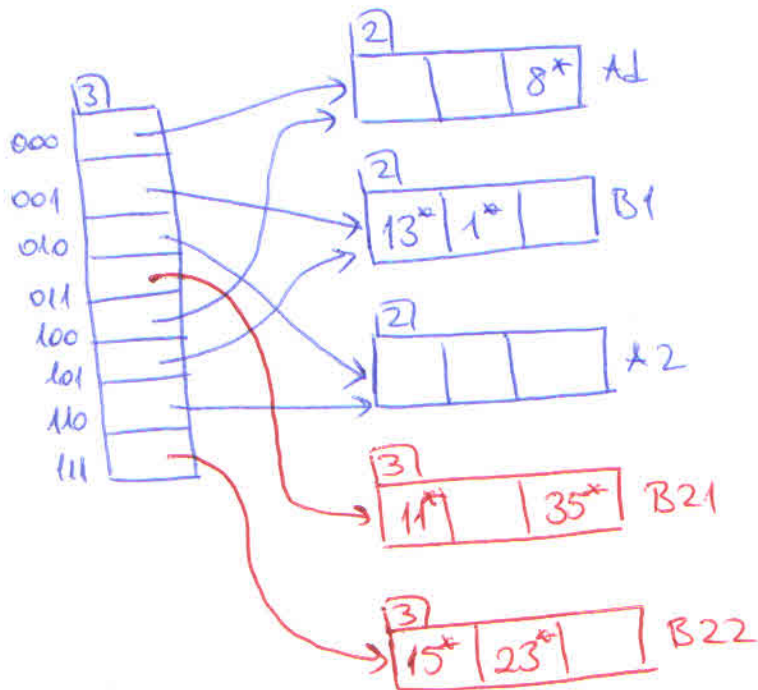
To insert 13, we need to split bucket B and double the directory as local depth of B = global depth.



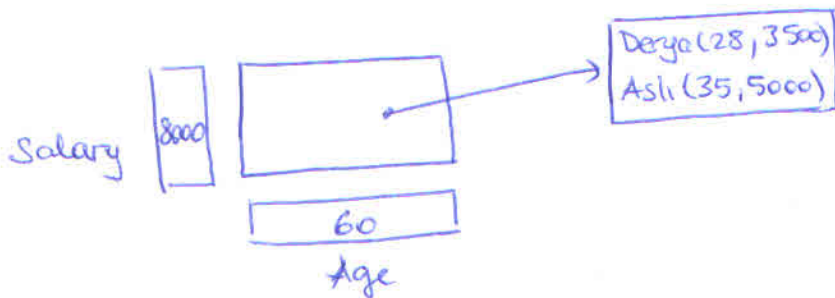
To insert 26, we need to split bucket A. As local depth of A < global depth, we don't double the size of the directory. We only split bucket A as follows:



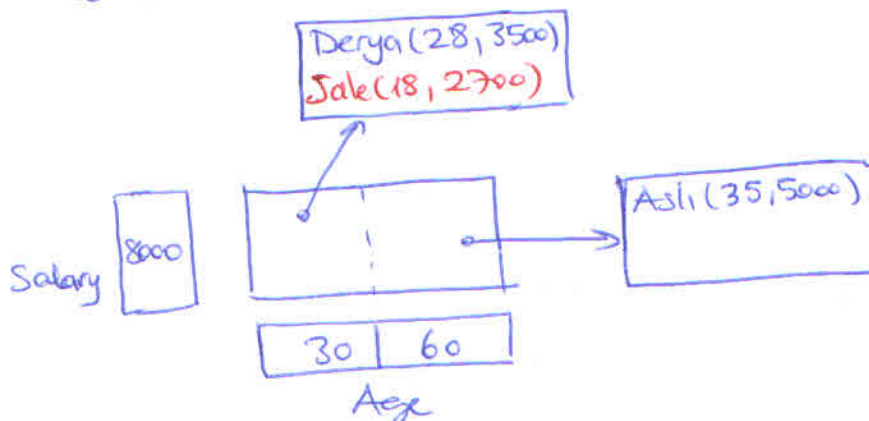
To insert 11, we need to split bucket B2, and double the size of the directory.



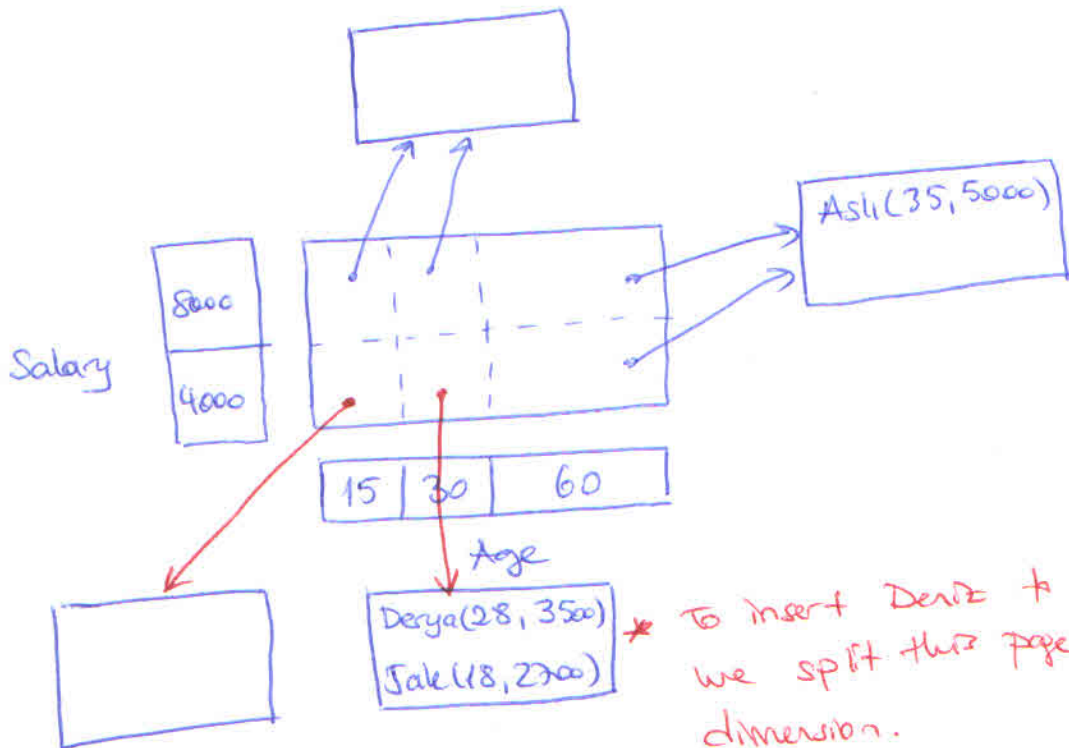
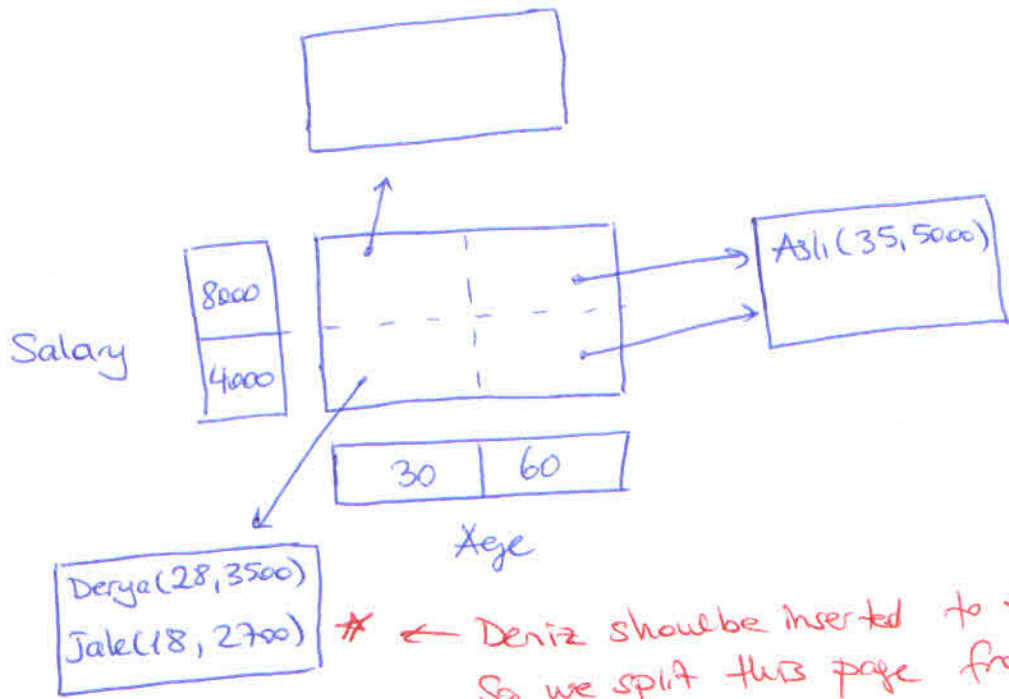
5. The first 2 records are inserted as follows:



To insert Jake, we split the grid from the Age dimension.



To insert Deniz (27, 3500) we split from Salary dimension.



Salary

8000
4000
2000



15	30	60
----	----	----

Age

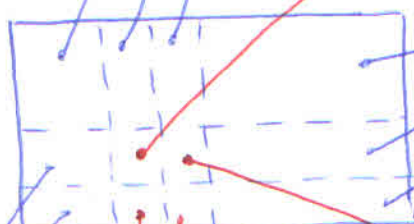
Asli(35, 5000)

Derya(28, 3500)
Jale(18, 2700)

★ To insert Deniz, we split this page from Age dimension.

Salary

8000
4000
2000



15	22	30	60
----	----	----	----

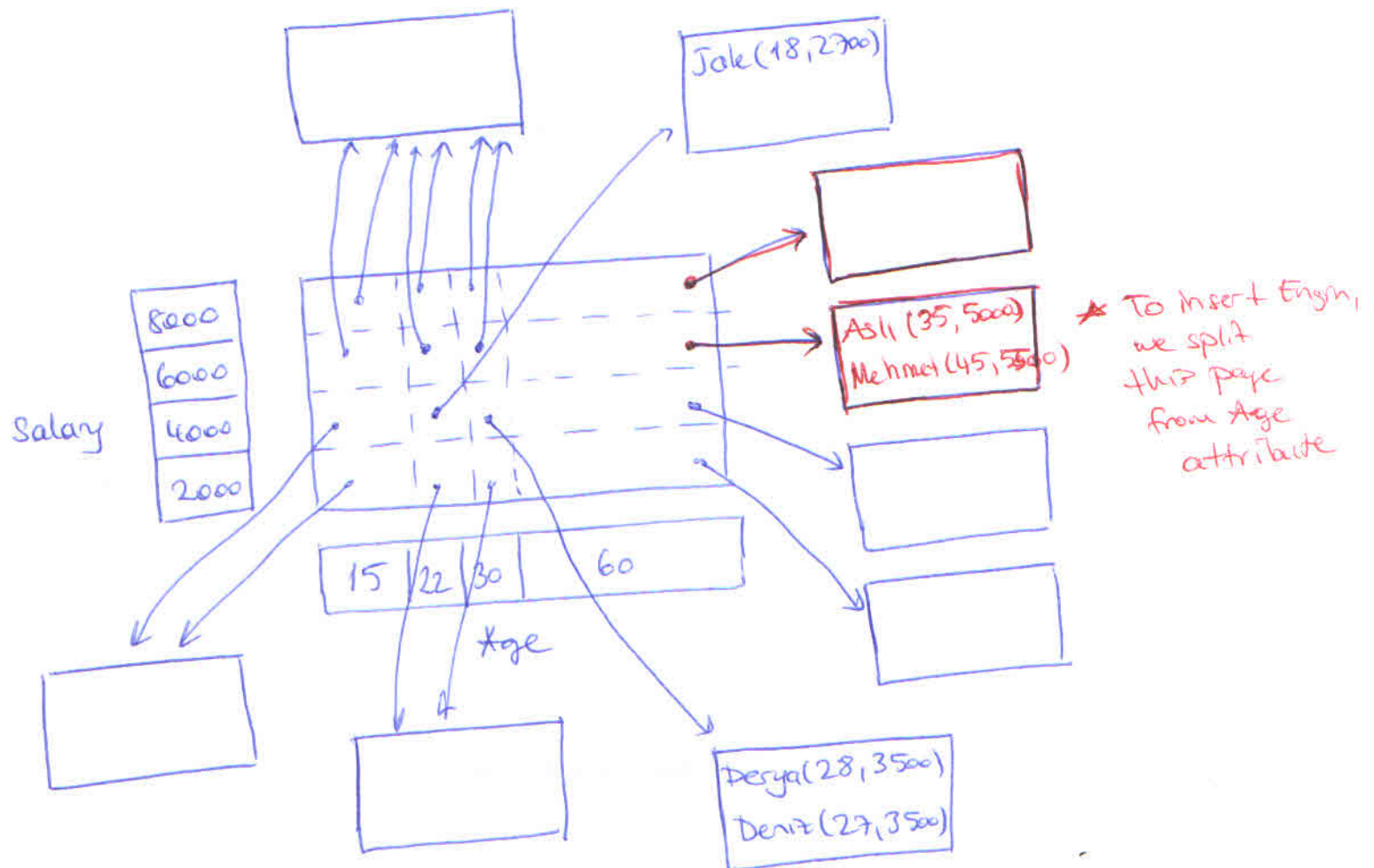
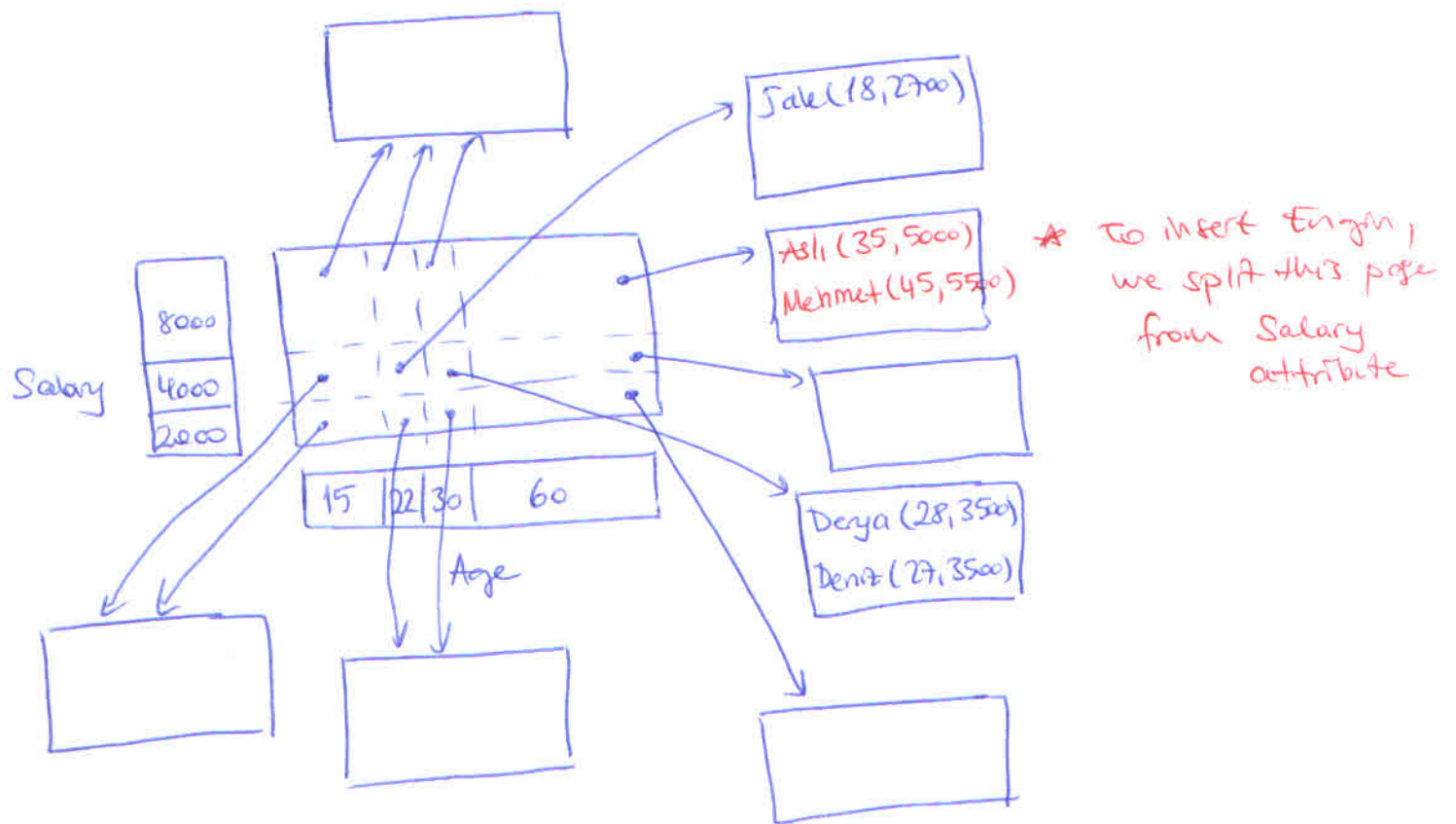
Age

Jale(18, 2700)

Asli(35, 5000)
Mehmet(45, 5500)

★ To insert Engin, we only split this page, not the grid.

Derya(28, 3500)
Deniz(27, 3500)



Salary

8000

6000

4000

2000

Jale (18, 2700)
Sami (19, 2600)

Aslı (35, 5000)
Mehmet (45, 5500)

Engin (55, 5500)
Rüya (50, 5500)

Derya (28, 3500)
Deniz (27, 3500)

15 22 30 45 60

Age

